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Neural Text Generation and Dialogue Modeling

① Natural Language Generation
② Neural Summarization
③ Reinforcement Learning
④ Dialogue Systems
⑤ Dialogue Challenges
⑥ Strategic Dialogue

Objectives

- Understand core tasks in **Neural Text Generation (NLG)**.
- Compare **extractive vs. abstractive summarization** and bottom-up methods.
- Explain why **Reinforcement Learning (RL)** is used in summarization and dialogue.
- Analyze key challenges in **Seq2Seq dialogue systems**.
- Understand strategy learning in **negotiation and strategic dialogue**.

Computer Understanding of
NATURAL LANGUAGE
(CAP 6640)
NLG

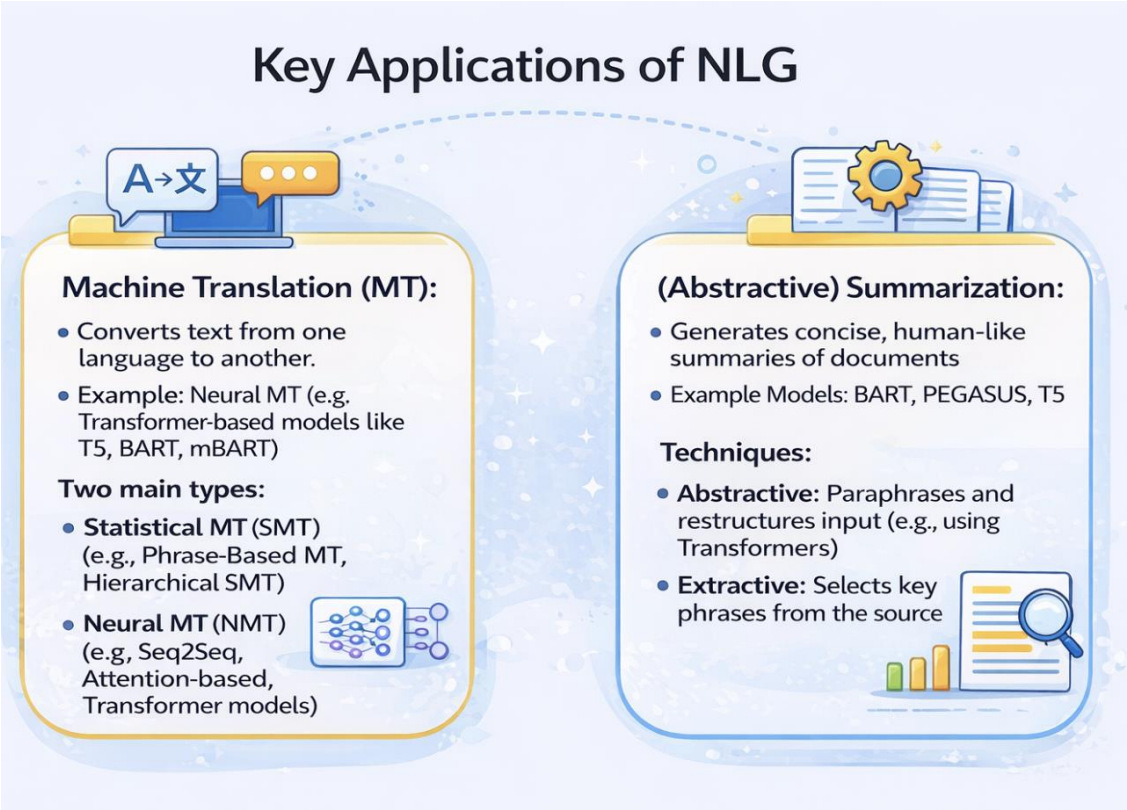
Natural Language Generation (NLG)

- NLG refers to the task of generating new text based on a given input (structured data, prompts, or unstructured sources).
  - It is a crucial subfield of NLP, enabling machines to produce coherent, contextually relevant, and human-like text.

- **Key Applications of NLG:**
  - Machine Translation (MT):
  - Abstractive Summarization

Natural Language Generation (NLG)

- **Key Applications of NLG:**
  1. **Machine Translation (MT):**
     - Converts text from one language to another.
     - **Example:** Neural MT (e.g., Transformer-based models like T5, BART, mBART).
     - Two main types:
       - **Statistical MT (SMT)** (e.g., Phrase-Based MT, Hierarchical SMT)
       - **Neural MT (NMT)** (e.g., Seq2Seq, Attention-based, Transformer models).
  2. **(Abstractive) Summarization:**
     - Generates concise, human-like summaries of documents.
     - **Example Models:** BART, PEGASUS, T5.
     - **Techniques:**
       - Abstractive: Paraphrases and restructures input (e.g., using Transformers).
       - Extractive: Selects key phrases from the source.



Natural Language Generation (NLG)

- **Key Applications of NLG (cont.)**
  3. **Dialogue Systems:**
     - **Chit-chat models:** Open-domain conversational agents (e.g., ChatGPT, BlenderBot).
     - **Task-oriented dialogue:** Goal-driven response generation (e.g., restaurant booking, customer support).
     - **Techniques:**
       - Retrieval-based vs. Generative models.
       - Reinforcement learning for dialogue optimization.
  4. **Creative Writing:**
     - **Storytelling & Poetry Generation:** AI-generated fiction, poetry, and artistic text.
     - **Example Models:** GPT-3/4, Bard, LLaMA.
     - **Challenges:** Style control, coherence, and consistency over long text spans.

Natural Language Generation (NLG)

- **Key Applications of NLG (cont.)**
  5. **Freeform Question Answering (QA):**
     - Generates answers from an open-ended prompt rather than selecting from predefined options.
     - **Example:** Open-domain QA (e.g., GPT-based models, RAG models).
     - **Techniques:** Retrieval-Augmented Generation (RAG), fine-tuning on QA datasets (SQuAD, HotpotQA).
  6. **Image Captioning:**
     - Generates descriptive captions for images.
     - **Model Architectures:** CNNs (for vision) + Transformers (for text) (e.g., CLIP, BLIP, Flamingo, SimVLM).
     - **Example:** COCO Captioning Challenge, Show-Attend-Tell model.

Summarization

Bottom-Up Summarization

- Bottom-up summarization is a method that **selectively extracts** salient content from the source text before generating an abstractive summary.
  - It combines **extractive and abstractive approaches**, where the model first determines key phrases and sentences, then paraphrases and reorganizes them for coherence.

- Advantages:

✅ Reduces hallucinations common in purely abstractive models.
✅ Balances informativeness and conciseness.
✅ Provides better content control over traditional end-to-end abstractive summarization.

Bottom-Up Summarization, cont.

- **Content Selection (Extractive Phase):**
  - Identifies important sentences or phrases using:
  - **Attention mechanisms** (e.g., self-attention in Transformers).
  - **Salience scoring models** (e.g., TF-IDF, BERT-based classifiers).
  - **Reinforcement learning** for importance ranking.
- **Abstractive Generation (Neural Rewriting):**
  - Uses an encoder-decoder model (e.g., Transformer, BART, T5) to rewrite selected content.
  - **Ensures coherence, fluency, and factual consistency** with the original text.

Examples

- Bottom-Up Attention (Gehrmann et al., 2018)
- BART-based controlled summarization
- Extract-and-Generate frameworks (e.g., FastRL)

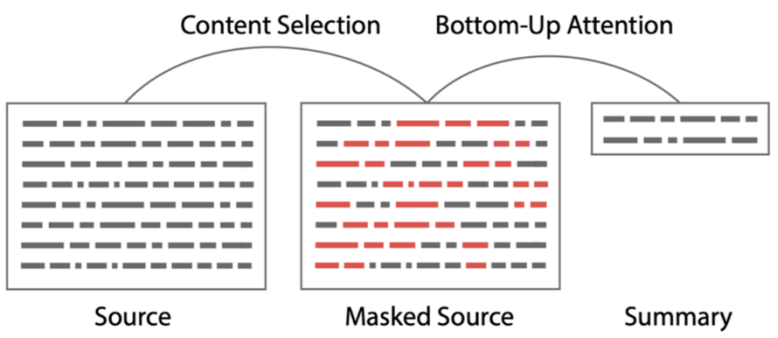


Figure 2: Overview of the selection and generation processes described throughout Section 4.

Simple but effective!

- Better overall content selection strategy
- Less copying of long sequences (i.e. more abstractive output)

RL-Based Summarization

- Neural summarization via **Reinforcement Learning (RL)** optimizes summary generation by **maximizing long-term rewards**, aligning model outputs with human evaluation metrics.
  - Overcomes limitations of standard sequence-to-sequence (Seq2Seq) models trained with Maximum Likelihood Estimation (MLE), which suffer from **exposure bias** and **mismatch between training and evaluation metrics** (e.g., ROUGE).

- **Advantages:**

✅ **Improved long-term coherence** by considering future word sequences.
✅ **Alignment with human judgments** via learned reward functions.
✅ **Reduction in exposure bias** compared to pure MLE training.

RL-Based Summarization, Why

- **MLE-based summarization issues:**
  - **Exposure bias:** The model only sees gold-standard summaries during training but must generate text during inference.
  - Mismatch between training (token-level) and evaluation (summary-level) objectives.
  - **Lack of diversity** in generated outputs.
- **RL provides a better optimization framework:**
  - Trains models to **directly optimize** evaluation metrics (e.g., ROUGE, METEOR, BLEU).
  - Uses a **reward function** to guide learning toward more human-preferred summaries.
  - Can incorporate **human feedback** (e.g., reward modeling from human preferences).

- **Challenges:**

❌ High variance in policy gradients leads to **unstable training**.
❌ **Defining a good reward function** is non-trivial.
❌ **Computationally expensive** compared to standard fine-tuning.

RL-Based Summarization, How

- **Policy Gradient Methods (REINFORCE)**
  1. Directly optimizes a reward function using **Monte Carlo sampling**.
  2. Example: **Self-Critical Sequence Training (SCST)** (Rennie et al., 2017), where the baseline is a greedy-decoded summary.
- **Actor-Critic Methods**
  1. Uses an **actor** (summary generator) and a **critic** (estimates the reward more accurately).
  2. Improves stability over pure policy gradient methods.
  3. Example: **Deep Reinforcement Learning with Advantage Actor-Critic (A2C)** for summarization.
- **Reward-Shaped Training**
  1. Hybrid models combining **MLE pretraining + RL fine-tuning** to improve stability.
  2. Example: **BERTSUM + RL**, **BART + RL fine-tuning**.

📌 **BERTSUM-RL** (Liu & Lapata, 2019) → Extractive + Abstractive hybrid with RL fine-tuning.
📌 **FastRL** (Chen et al., 2022) → Efficient RL-based summarization with content weighting.
📌 **Human Feedback-Based Summarization** (Stiennon et al., 2020) → Uses human pref ranking for reward modeling.

RL-Based Summarization, Results

- Interesting findings: using RL instead of ML achieved higher ROUGE scores, but lower human judgement

Model	ROUGE-1	ROUGE-2	ROUGE-L
ML, no intra-attention	44.26	27.43	40.41
ML, with intra-attention	43.86	27.10	40.11
RL, no intra-attention	**47.22**	30.51	**43.27**
ML+RL, no intra-attention	47.03	**30.72**	43.10

Model	Readability	Relevance
ML	6.76	7.14
RL	4.18	6.32
ML+RL	**7.04**	**7.45**

❌ "We observed that our models with the highest ROUGE scores also generated barely-readable summaries."

✅ Overall, hybrid approach does best.

Dialogue

Dialogue

- **Dialogue systems** refer to NLP models that process and generate **interactive text-based conversations** with users.
  - These systems can be broadly categorized into **task-oriented** and **social dialogue** models.

- **Types of Dialogue**
  - Task-oriented dialogue
    - Assistive, cooperative, adversarial dialogue
  - Social dialogue
    - Chit-chat dialogue
    - Therapeutic dialogue

Evolution of Dialogue Systems

- **Before 2015: Rule-Based & Retrieval-Based Dialogue**
  - Open-ended free NLG was difficult due to:
    - Lack of robust language modeling techniques
    - Limited training data for generative models
  - Most dialogue systems relied on:
    - Predefined templates (e.g., handcrafted rule-based systems).
    - Retrieval-based methods (selecting responses from a fixed corpus).
    - Example: ELIZA (1966), A.L.I.C.E (1995), early chatbot systems.

- **Post-2015: Seq2Seq & Neural-Based Dialogue Systems**
  - Breakthrough in neural sequence-to-sequence (Seq2Seq) modeling with:
    - Recurrent Neural Networks (RNNs) & LSTMs for response generation.
    - Attention mechanisms improving context retention.
    - Example: Google's Neural Conversational Model (Vinyals & Le, 2015).

Dialogue Types

- **Task-Oriented Dialogue:** Aimed at achieving a specific goal by assisting users in completing tasks.
  - **Assistive Dialogue:** Customer service bots, recommender systems, personal assistants.
    - Ex: ChatGPT, Google Assistant, Alexa, Siri.
    - Uses goal-driven, slot-filling architectures (e.g., Dialogue State Tracking).
  - **Cooperative Dialogue:** Two agents work together to solve a task.
    - Ex: Multi-agent collab. planning (e.g., AI teammates in games, negotiation bots).
    - Techniques: **Reinforcement Learning, Hierarchical Dialog Policies**.
  - **Adversarial Dialogue:** Two agents **compete** in a conversation.
    - Ex: Debate models (e.g., IBM Project Debater), negotiation bots, persuasion bots.
    - Uses game theory-based approaches, RL, and strategic dialogue planning.

Dialogue Types, cont.

- **Social Dialogue:** Designed for **engagement, companionship, or emotional support** rather than achieving a specific task.
  - **Chit-Chat:** Free-form conversation for fun or casual interaction.
    - Ex: BlenderBot, ChatGPT, LaMDA.
    - Uses Transformer-based models trained on large-scale dialogue datasets (Reddit, Twitter, Persona-Chat).
  - **Therapeutic Dialogue:** AI-driven therapy and mental well-being applications.
    - Ex: Woebot, Replika, Wysa.
    - Uses sentiment-aware NLP, empathetic response generation, and cognitive behavioral therapy (CBT) principles.

Seq2Seq-Based Dialogue Systems

- While early **Seq2Seq + Attention** models improved dialogue generation, they exhibited **severe deficiencies** in open-domain **chit-chat**:
  - Generic and Boring Responses
  - Irrelevant Responses
  - Repetition
  - Lack of Context Retention
  - Lack of Consistent Persona

Seq2Seq-Based Dialogue Systems Issues

- **Generic & Boring Responses**
  - Models often generate high-probability yet **uninformative** responses.
  - Example: *"I don't know"* or *"That's interesting"*.
  - **Why?** Training loss optimizes for safe, high-likelihood responses.
- **Irrelevant Responses**
  - Generated replies **drift from the conversation context**.
  - **Why?** Encoder **fails to capture nuanced dependencies** in multi-turn dialogue.
- **Repetition**
  - Tendency to **repeat phrases** within or across responses.
  - **Why?** Models overfit on high-frequency patterns in the dataset.

Seq2Seq-Based Dialogue Systems Issues

- **Lack of Context Retention**
  - Responses ignore **previous turns** in conversation.
  - **Why?** Seq2Seq models **lack explicit memory** of past exchanges.

- **Lack of Consistent Persona**
  - Models generate **contradictory or inconsistent** statements about themselves.

  - **Why?** No persistent user or AI **personality embedding** in standard architectures.

Response Genericness & Boring Dialogue

- **Easy Test-Time Fixes**
  - **Upweight Rare Words** in **Beam Search:** Encourages more diverse responses by boosting low-frequency terms.
  - **Use Sampling-Based Decoding:** Instead of selecting the highest probability token (greedy/beam search), use:
    - **Top-k Sampling:** Restrict selection to the k most probable words.
    - **Nucleus Sampling (Top-p):** Sample from a **probability mass p** rather than fixed *k*.
- **Conditioning Fixes**
  - **Condition the Decoder on Additional Content:** Extract key **content words** and explicitly **attend to them** during decoding.
  - **Retrieve-and-Refine Model Instead of Pure Generation**
    - First, **retrieve** a human-written utterance from a corpus.
    - Then, **edit** it to fit the current conversation context.
    - Produces **more diverse, human-like responses**.

Dialogue Issues

Irrelevant Response in Dialogue

- **Problem: Irrelevant Responses in Seq2Seq Dialogue Models**
  - Standard Seq2Seq models maximize **P(T | S)** (likelihood of response T given input S).
  - Leads to **generic, contextually weak, or irrelevant responses**.
- **Solution: Optimizing for Maximum Mutual Information (MMI)**
  - Instead of just maximizing **P(T | S)**, MMI introduces:
    - **P(S | T):** Ensures the response T is also likely to generate input S.
    - **P(T):** Helps **penalize generic responses** by considering response frequency.
  - **MMI Objective Function:**

$$\hat{T} = \arg\max_{T} \left[\log P(T|S) - \lambda \log P(T) \right]$$

Ensures response **T is relevant** given input **S**

Penalizes **generic** high-frequency responses

Repetition in Dialogue Systems

- **Problem: Repetitive Responses in Dialogue Generation: Seq2Seq models** often generate repetitive phrases due to:
  - **Over-reliance on high-probability tokens**.
  - **Lack of explicit mechanisms to track previously generated content**.
  - **Exposure bias** during training (model sees only ground-truth sequences, not its own generated outputs).
- **Simple Fix: Blocking n-gram Repetitions**
  - **Directly prevent repeating n-grams during beam search**.
  - Example: If *"I think that"* has been generated, block generating it again.
  - **Easy to implement** and often **highly effective**.

Advanced Solutions for Repetition

- **Train a Coverage Mechanism**
  - Inspired by coverage models in machine translation.
  - Prevents attention from focusing on the same words multiple times.
  - Ex: Coverage vector tracks words already attended.
- **Define a Training Objective to Discourage Repetition**
  - Modify loss function to penalize repeated tokens.
  - Ex: Diversity-promoting loss functions (e.g., mutual information objectives).
- **RL for Non-Differentiable Repetition Control**
  - If repetition penalty is a non-differentiable function, use RL techniques.
  - Reward-based learning to promote diverse, natural conversations.
  - Example: Self-Critical Sequence Training (SCST) with diversity-based rewards.

✅ **Improves response diversity** and reduces unnatural loops.
✅ **Encourages more engaging, human-like conversations**.
✅ **Combines well with other techniques like MMI and persona modeling**.

Lack of Consistent Persona in Dialogue

- **Problem:** Standard Seq2Seq models generate **incoherent, contradictory responses** about the AI's persona.
  - Models lack **long-term memory** or **personalization mechanisms**.
  - Example:
    - *User:* "Where are you from?" → *AI:* "I'm from New York."
    - *Later in conversation:* "I live in London."

- **Early Solution: Persona-Embedded Seq2Seq Model (Li et al., 2016)**
  - **Approach:** Encodes **both conversation partners' personas** into **latent embeddings**. Conditions **response generation** on persona embeddings.
  - **Impact:** Ensures **response remain consistent** across multi-turn dialogues. Provides a foundation for **persona-aware models**.

Lack of Consistent Persona in Dialogue

- **Modern Solution: PersonaChat (Zhang et al., 2018)** → A large-scale dataset for **persona-driven dialogue**.
  - Each **persona** is a **set of 5 sentences** describing personal traits. Ex.
    - *"I love hiking."*
    - *"I have a pet dog."*
    - *"I enjoy reading sci-fi books."*
  - **How It Helps:**
    - Provides **light grounding** for **training persona-aware dialogue agents**.
    - Enables models to **maintain personality consistency** across conversations.

Negotiation Dialogue

Negotiation Dialogue in NLP

- **Lewis et al. (2017) - Negotiation Dataset**
  - Two agents **negotiate** via **natural language** to divide items.
  - Agents have **different value functions** for each item.
  - Dialogue continues until they **reach an agreement**.
- **Challenges:**
  - **Strategic reasoning:** Understanding opponent's preferences.
  - **Deception & persuasion:** Generating effective negotiation tactics.
  - **Language ambiguity:** Handling indirect or implicit offers.
- **Applications:**
  - AI-driven contract negotiation
  - Game AI for multi-agent decision-making
  - Human-AI collaboration in bargaining

RL for Negotiation Dialogue

- Seq2Seq models trained with Maximum Likelihood (ML) produce fluent but weak strategic negotiators.
- Inspired by Paulus et al. (2018) in summarization, RL is introduced to optimize for a discrete reward.
- Agents train by playing against themselves, improving negotiation tactics.
- **Training Approach:**
  - **Reinforcement Learning (RL) Objective:** Optimizes negotiation success.
  - **Combined with ML Objective:** Maintains fluency and coherence.
- **Potential Pitfall:**
  - Pure RL optimization can lead to agents diverging from natural English, producing gibberish.

RL for Negotiation Dialogue

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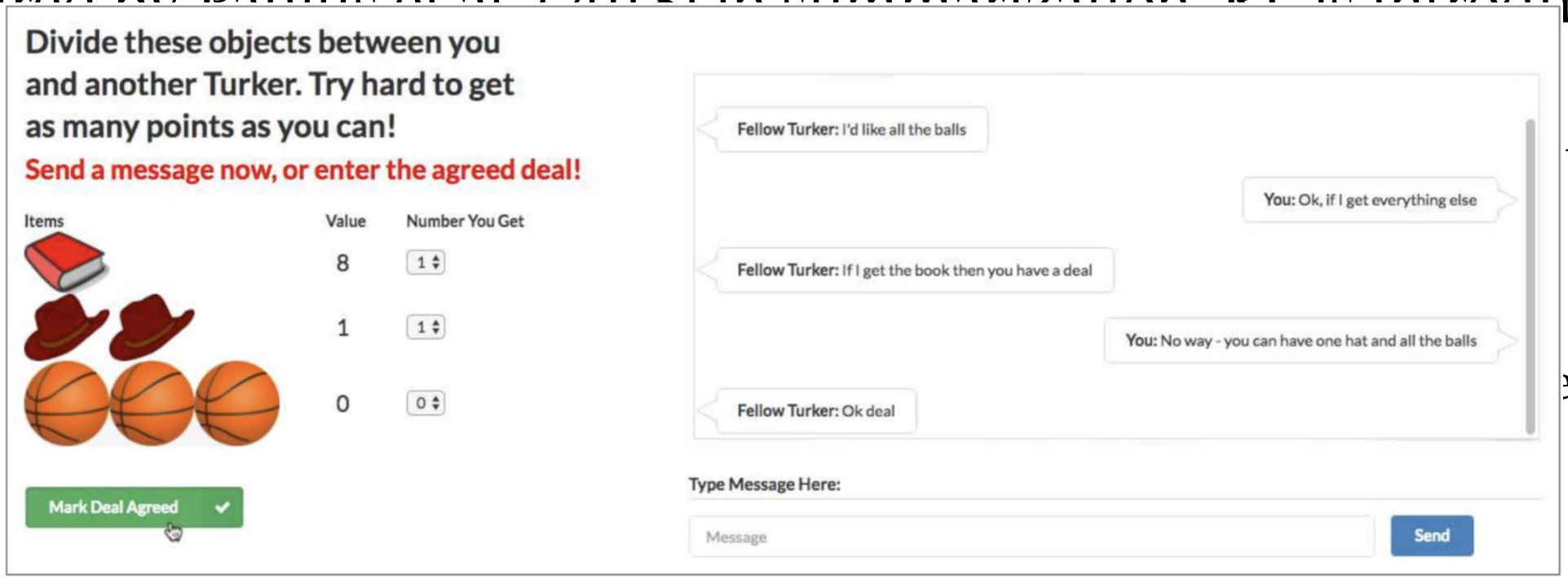

Strategic-NLG Separation in Dialogue

- **Idea:** Separates strategy from natural language generation (NLG) in negotiation dialogue.

- **Approach:**
  - Each utterance x_t has a discrete latent variable z_t.
  - z_t is trained to predict future events (e.g., upcoming messages, strategic outcome).
  - However, z_t does not predict x_t itself.

- **Why this matters:**
  - z_t captures dialogue impact, not surface wording.
  - Enhances controllability and interpretability.
  - Makes strategy learning easier and more structured.

Strategic-NLG Separation in Dialogue

- Model evaluates **possible responses** by **simulating future dialogue** (rollouts).
  - It selects the response that **maximizes expected reward** over the predicted conversation flow.
- **How:**
  - **Generate candidate responses** for the current turn.
  - **Simulate future dialogue** for each candidate.
  - **Compute expected reward** (e.g., negotiation success, coherence).
  - **Choose the response** with the **highest expected reward**.

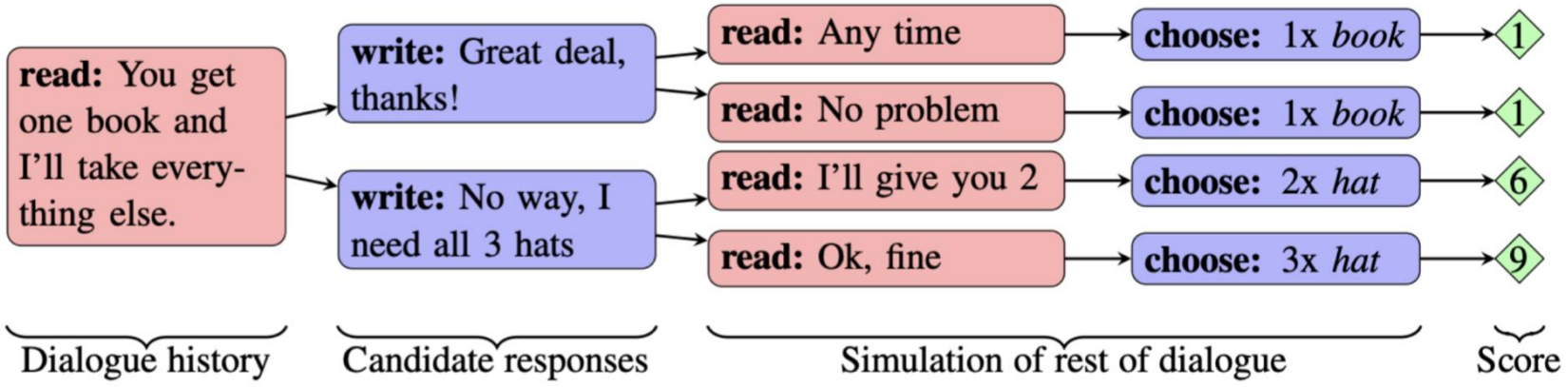

Figure 4: Decoding through rollouts: The model first generates a small set of candidate responses. For each candidate it simulates the future conversation by sampling, and estimates the expected future reward by averaging the scores. The system outputs the candidate with the highest expected reward.

Key Takeaways

- NLG spans MT, summarization, dialogue, QA, and multimodal generation.
- Pure MLE training causes exposure bias and generic responses.
- RL aligns generation with long-term objectives but introduces instability.
- Dialogue systems suffer from genericness, repetition, and persona inconsistency.
- Strategic dialogue separates **planning (strategy)** from **surface realization (NLG)**.